If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

a. All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.

b. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.

c. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.

d. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).

e. Bonus problems are not required for credit, but they are worth extra credit.

Definitions & Facts. Recall that for a linear system of the form $Ax = b$,

- There exists a solution if $b \in \mathrm{Range}(A)$.
- The solution is unique if A has full column rank, and in particular, the least squares solution is given by $x = (A^\top A)^{-1} A^\top b$.
- If A does not have full column rank, then there are infinitely many solutions.

Problems

Problem 1. (Casting a problem as a least squares problem.) Consider the data points $(x_i, y_i) \in \mathbb{R}\times\mathbb{R}$:

$$\{(x_i, y_i)\}_{i=1}^4 = \{(1,1), (2,0), (-1,2), (0,-1)\}.$$

We aim to determine a real polynomial of degree two that best fits this data, where the polynomial is of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$, and $\theta = (\theta_0, \theta_1, \theta_2)^\top \in \mathbb{R}^3$. Note that there are more data points that there are unknown coefficients, so this problem is overdetermined.

(a) Write the objective function as a sum of norms defined in terms of the coefficients.

(b) Write out the matrix-vector equation that defines the problem (i.e., what is X and y?)

(c) Solve the problem using the normal equations.

Problem 2. (Definition and basic computation.)
Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

(a) Write down the definition of the characteristic polynomial of a square matrix A.

(b) Compute the characteristic polynomial $p_A(\lambda)$ of A.

(c) Find the eigenvalues of A.

Problem 3. (Characteristic polynomial and trace/determinant.)
Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

(a) Compute the characteristic polynomial $p_A(\lambda)$ symbolically.

(b) Express $p_A(\lambda)$ in terms of $\mathrm{tr}(A)$ and $\det(A)$.

(c) Verify that the sum of eigenvalues equals $\mathrm{tr}(A)$ and the product equals $\det(A)$.

Problem 4. ((VMLS 12.15) Minimizing a squared norm plus an affine function.)
Consider

$$\min_{x \in \mathbb{R}^n} \quad \|Ax - b\|^2 + c^\top x + d,$$

where $A \in \mathbb{R}^{m \times n}$ has linearly independent columns, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, $d \in \mathbb{R}$.
Show that the objective can be written

$$\|Ax - b\|^2 + c^\top x + d \;=\; \|Ax - b + f\|^2 + g$$

for some $f \in \mathbb{R}^m$ and constant $g \in \mathbb{R}$, and identify a valid choice of f and g. Hence the problem reduces to an ordinary least–squares problem.
Hint: Express the normal squared term on the right-hand side as $\|(Ax - b) + f\|^2$ and expand. Then argue the equality above holds provided that $2A^\top f = c$. One possible choice is $f = \frac{1}{2}\,(A^\dagger)^\top c$. (You must verify these steps.)

Problem 5. (Projection Matrix onto a Subspace.) Let $X \in \mathbb{R}^{m \times n}$ have linearly independent columns, representing n basis vectors for a subspace

$$\mathcal{C} := \mathrm{col}(X) = \{z \mid \exists \theta \in \mathbb{R}^n, z = X\theta\} \subseteq \mathbb{R}^m.$$

For any vector $y \in \mathbb{R}^m$, we define its projection onto $\mathcal{C}$ as

$$\hat{y} := Py, \qquad \text{where } P = X(X^\mathsf{T} X)^{-1} X^\mathsf{T}.$$

(a) Show that P is symmetric and idempotent; that is,

$$P^\mathsf{T} = P, \qquad P^2 = P.$$

(b) Show that $\hat{y} = Py$ is the unique vector in $\mathrm{col}(X)$ that minimizes

$$\|y - X\theta\|_2^2.$$

(Hint: use the normal equations $X^\mathsf{T} X \hat{\theta} = X^\mathsf{T} y$.)

(c) Verify that the residual $r := y - \hat{y}$ satisfies $X^\mathsf{T} r = 0$.

(d) Let

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad y = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

Compute P, $\hat{y}$, and the residual $r = y - \hat{y}$ explicitly. Check that r is orthogonal to the columns of X.

Problem 6. (Least Squares and QR Factorization.) Suppose X is an $m \times n$ matrix with linearly independent columns, and $X = QR$ is its QR factorization, where $Q \in \mathbb{R}^{m \times n}$ has orthonormal columns ($Q^\mathsf{T} Q = I_n$), and $R \in \mathbb{R}^{n \times n}$ is upper triangular. For a given vector $b \in \mathbb{R}^m$, the least-squares solution $\hat{\theta}$ minimizes $\|X\theta - y\|_2$. The vector $X\hat{\theta}$ is the projection of y onto the column space of X.

(a) Show that $X\hat{\theta} = QQ^\mathsf{T} y$. (The matrix QQ^T is called the projection matrix.)

(b) Show that

$$\|X\hat{\theta} - y\|_2^2 = \|y\|_2^2 - \|Q^\mathsf{T} y\|_2^2.$$

(This is the square of the distance between y and the closest linear combination of the columns of X.)

Problem 7. (Minimum–Norm Trick for Linear Systems.) We want to find the smallest–norm vector θ that satisfies the linear system

$$X\theta = y,$$

where $X \in \mathbb{R}^{n \times d}$ has full row rank ($n < d$) and the system is consistent. This problem is underdetermined and can be cast as the following optimization problem:

$$\min_{\theta} \|\theta\|_2 \quad \text{s.t. } X\theta = y.$$

(a) Instead of solving the constrained problem directly, consider the *augmented objective*

$$J(\theta) = \tfrac{1}{2}\|\theta\|_2^2 + \lambda^\mathsf{T}(y - X\theta),$$

where $\lambda \in \mathbb{R}^m$ is a fixed vector.

Compute $\nabla_\theta J(\theta)$ and find θ that minimizes $J(\theta)$ in terms of λ.

(b) Substitute your result into the constraint $X\theta = y$ and solve for λ in terms of X and y.

(c) Substitute λ back to get an explicit expression for θ. Show that

$$\hat{\theta} = X^\mathsf{T}(XX^\mathsf{T})^{-1} y.$$

(d) Explain why this $\hat{\theta}$ is the same as the solution to

$$\min_{\theta} \|\theta\|_2 \quad \text{s.t. } X\theta = y.$$

(Hint: do this in steps. First, first consider any θ that satisfies $X\theta = y$. Then, show that θ can be written as $\theta = \theta_0 + v$, where θ_0 is a particular solution and v is in the null space of X. Finally, by expanding the norm $\|\theta\|_2^2 = \|\theta_0 + v\|_2^2$, show that the minimum norm solution is the one that minimizes $\|\theta\|_2$ over all θ that satisfy $X\theta = y$.)

(e) Let

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Compute $\hat{\theta}$ explicitly and check that $X\hat{\theta} = y$ and that $\hat{\theta}$ has the smallest norm among all feasible solutions.